

MH1200 Linear Algebra I.

Solutions to the Problem list for Week #1.

Problem 1:

Find the general solutions of the following linear equations

$$(i) \quad x + y + z = 1 \qquad (ii) \quad x - y + 3z = 5$$

What is the geometric interpretation of the solution set in each case?

Solution to (i).

We can set y and z to anything at all, and then solve for an x that makes the equation true for that y and z . (Actually we could choose any two of the three variables here, but it is traditional to choose these two here. You will understand why this is normal later.)

So we introduce two free parameters $r, s \in \mathbb{R}$ and set $y = r$ and $z = s$.

When we substitute these two assignments we get:

$$x + r + s = 1.$$

There is an x which makes this equation true no matter what r and s are and we solve for it:

$$x = 1 - r - s.$$

The corresponding set of solutions can be written:

$$\begin{cases} x = 1 - r - s \\ y = r \\ z = s \end{cases}, \quad r, s \in \mathbb{R}.$$

This is the full set of solutions because any specific solution (x', y', z') of the equation is obtained from this parametrization by setting $r = y'$ and $s = z'$.

To understand the geometry of this set of solutions let's start by rewriting it in vector format. The corresponding vector expression for the set of solutions is

$$S = \{(1 - r - s, r, s) \in \mathbb{R}^3, r, s \in \mathbb{R}\}.$$

To understand the geometric picture we rewrite this expression so it is decomposed in terms of elementary geometric operations. We do it like this:

$$(1 - r - s, r, s) = (1, 0, 0) + (-r, r, 0) + (-s, 0, s) = (1, 0, 0) + r(-1, 1, 0) + s(-1, 0, 1).$$

Thus the set we which to understand is:

$$S = \{(1, 0, 0) + r(-1, 1, 0) + s(-1, 0, 1) \in \mathbb{R}^3, r, s \in \mathbb{R}\}.$$

What is the picture here? If we forget the first term $(1, 0, 0)$ the simpler set

$$\{r(-1, 1, 0) + s(-1, 0, 1) \in \mathbb{R}^3, r, s \in \mathbb{R}\}$$

is the plane containing the origin and containing the arrows $(-1, 1, 0)$ and $(-1, 0, 1)$.

When we add the first term $(1, 0, 0)$ back in the effect is to shift that plane by 1 unit in the x -direction. (See also problem 8 for a deeper look at this step.)

Solution to (ii).

We will be a bit quicker with this case. The equation is: $x - y + 3z = 5$.

As before we introduce free parameters $r, s \in \mathbb{R}$, set $y = r$ and $z = s$, and solve

$$x = 5 + r - 3s.$$

The full set of solutions is thus:

$$\begin{cases} x &= 5 + r - 3s \\ y &= r \\ z &= s \end{cases}, \quad r, s \in \mathbb{R}.$$

Now to see the geometric picture we rewrite this in vector format as:

$$S = \{(5 + r - 3s, r, s) \in \mathbb{R}^3, r, s \in \mathbb{R}\}.$$

Rewriting this to decompose the formula into the two basic operations we get:

$$S = \{(5, 0, 0) + r(1, 1, 0) + s(-3, 0, 1) \in \mathbb{R}^3, r, s \in \mathbb{R}\}.$$

The geometric interpretation is that we take the plane through the origin containing the vectors $(1, 1, 0)$ and $(-3, 0, 1)$ and we translate it by 5 units in the x -direction.

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Problem 2:

Find the general solution of the following linear system. (In other words find the points (x, y, z) which solve both equations simultaneously.)

$$\begin{aligned} x + y + z &= 1 \\ x - y + 3z &= 5 \end{aligned}$$

Solution:

Let's try and follow the hint and see where we get. Assume we have some solution (x, y, z) . Then the first equation tells us that the variables are related by

$$x = 1 - y - z.$$

Then we can substitute this relationship into the second equation and we deduce that any solution of the system must satisfy the equation

$$(1 - y - z) - y + 3z = 5.$$

Simplifying this:

$$-2y + 2z = 4.$$

This is an equation we can solve completely. We introduce a free parameter $r \in \mathbb{R}$ and set $z = r$. Then we solve this to get $y = r - 2$.

If we substitute $y = r - 2$ and $z = r$ back into equation 1 we deduce that to be a solution x must be $x = 1 - r + 2 - r = 3 - 2r$.

It follows from the way we solved these equations that all of these triples

$$\begin{cases} x &= 3 - 2r \\ y &= r - 2 \\ z &= r \end{cases}, \quad r, s \in \mathbb{R}.$$

solve both of the two given equations (why, exactly?). (It is easy to check this, so please double-check it by substituting these formulae back into the equations.)

Next, we wonder what is the geometric picture? If we write this set of solutions in vector notation as:

$$\{(3 - 2r, r - 2, r) \in \mathbb{R}^3, r \in \mathbb{R}\}$$

then by decomposing into fundamental operations:

$$\{(3, -2, 0) + r(-2, 1, 1) \in \mathbb{R}^3, r \in \mathbb{R}\}$$

we see that this set is the line in $\mathbb{R}^3$ containing the point $(3, -2, 0)$ and in the direction $(-2, 1, 1)$.

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Problem 3:

Give a parametrization for the set of solutions for the following linear equation:

$$x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 6.$$

Solution

Notice that we can pick any 4 of these variables to be anything at all, and then the remaining variable is determined by the equation. It is traditional to set x_2, x_3, x_4 and x_5 to be free variables:

$$x_2 = r, \quad x_3 = s, \quad x_4 = t, \quad x_5 = u, \quad r, s, t, u \in \mathbb{R}$$

and then solve for

$$x_1 = 6 - 2r - 3s - 4t - 5u.$$

The parametrization for the set of solutions is

$$\begin{cases} x_1 &= 6 - 2r - 3s - 4t - 5u \\ x_2 &= r \\ x_3 &= s \\ x_4 &= t \\ x_5 &= u \end{cases}, \quad r, s, t, u \in \mathbb{R}.$$

We don't really have the language for describing the geometry yet. Later we might call this "A 4-dimensional affine subspace of $\mathbb{R}^5$ containing the point $(6, 0, 0, 0, 0)$ and the containing the directions $(-2, 1, 0, 0, 0)$, $(-3, 0, 1, 0, 0)$, $(-4, 0, 0, 1, 0)$, $(-5, 0, 0, 0, 1)$."

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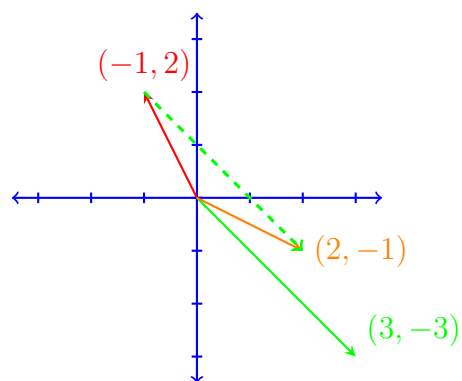
Problem 4:

Draw the following vectors:

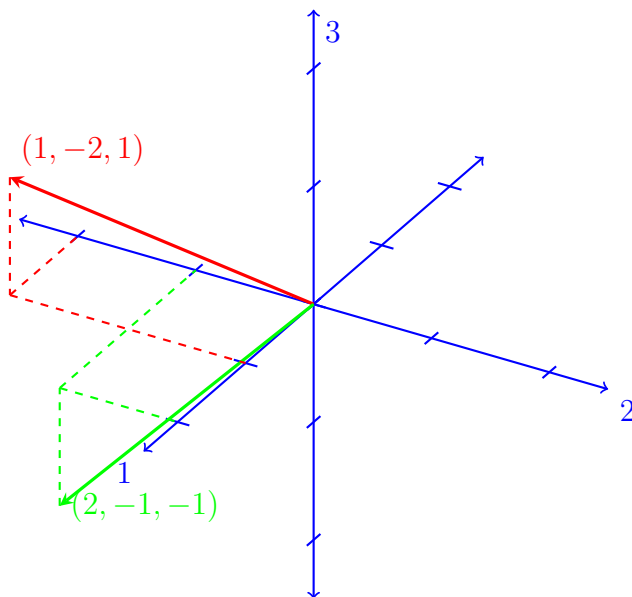
1. $(-1, 2)$
2. $(-1, 2) + (3, -3)$
3. $(1, -2, 1)$
4. $(2, -1, -1)$

Solution

Here is a picture for parts 1 and 2.



Here is a picture for parts 3 and 4.



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Problem 5:

Do there exist $a, b \in \mathbb{R}$ such that $(1, 1, 1) = a \cdot (1, -2, 1) + b \cdot (2, -1, -1)$?

Solution

An equation between vectors like this is equivalent to the system of linear equations you get by equating corresponding entries. Start by combining the terms on the RHS into a single vector:

$$(1, 1, 1) = a \cdot (1, -2, 1) + b \cdot (2, -1, -1) = (a + 2b, -2a - b, a - b).$$

Then we equate corresponding entries we get a system of three equations (one equation for each entry in the vector):

$$\begin{aligned} a + 2b &= 1 \\ -2a - b &= 1 \\ a - b &= 1 \end{aligned}$$

Let's try to solve these equations and see what happens. If we take twice the first equation and add it to the second equation we deduce: $3b = 3$, thus we deduce $b = 1$.

On the other hand if we add (-1) times the first equation to the third equation we deduce $-3b = 0$, thus $b = 0$.

These two deductions contradict each other. Thus there are no a and b which makes the vector equation true.

(There is an alternative nice solution you may have spotted: when you add up all three equations the LHSs cancel to 0, but the RHS adds up to 3. Thus if there was a solution it would imply that $0 = 3$ which is an impossibility.)

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Problem 6:

Let $\vec{u} = (1, 2)$ and $\vec{v} = (2, 1)$. Draw the vectors

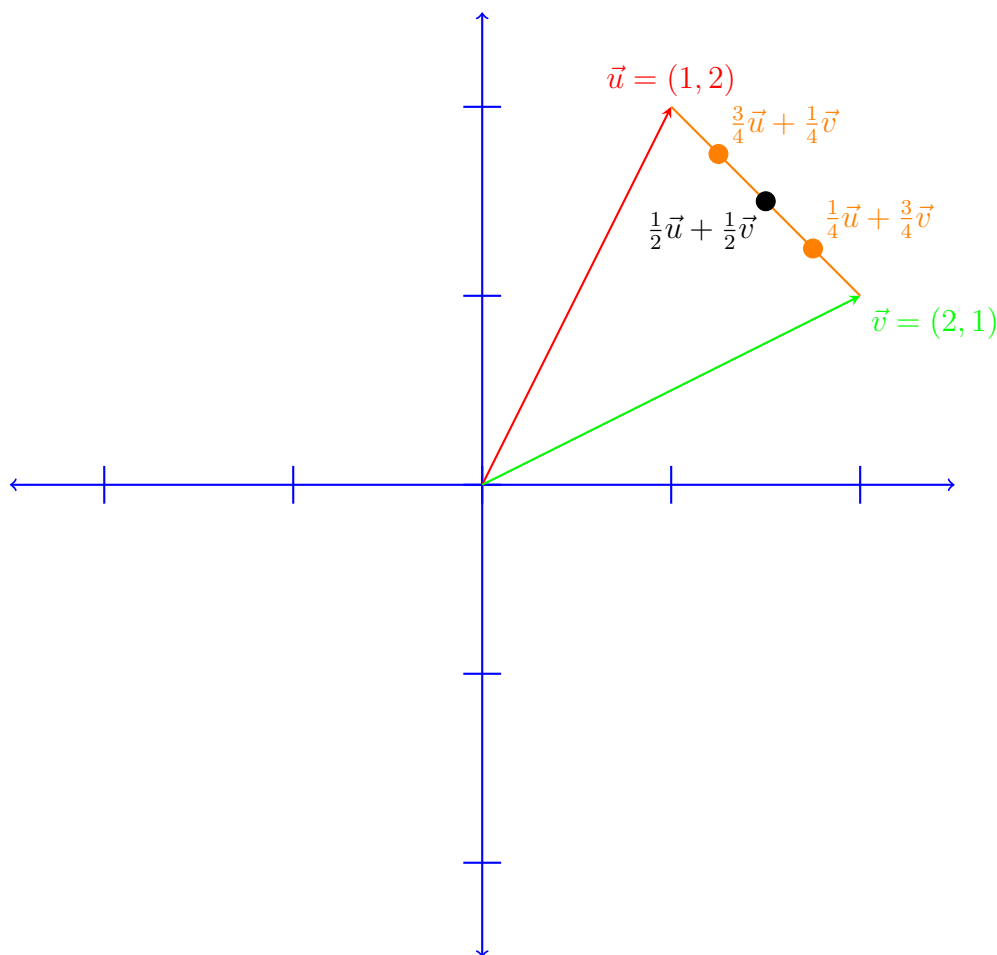
1. $\frac{1}{2}\vec{u} + \frac{1}{2}\vec{v}$
2. $\frac{3}{4}\vec{u} + \frac{1}{4}\vec{v}$
3. $\frac{1}{4}\vec{u} + \frac{3}{4}\vec{v}$

Solution

Here is the picture. Hopefully from these three points you can visualize all the points

$$c \cdot \vec{u} + (1 - c) \cdot \vec{v}$$

for $0 \leq c \leq 1$, which is part of Problem 7.



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Problem 7:

Let $\vec{u} = (1, 2)$ and $\vec{v} = (2, 1)$. Draw the following sets of vectors.

1. $\{c \cdot \vec{u} + (1 - c) \cdot \vec{v} : c \in \mathbb{R}, c \geq 0\}$.
2. $\{a\vec{u} + b\vec{v} : 0 \leq a \leq 1, 0 \leq b \leq 1\}$.
3. $\{a\vec{u} + b\vec{v} : a + b \leq 1\}$.

Solution

1. The orange ray. One trick to see this clearly is to rewrite the expression as follows:

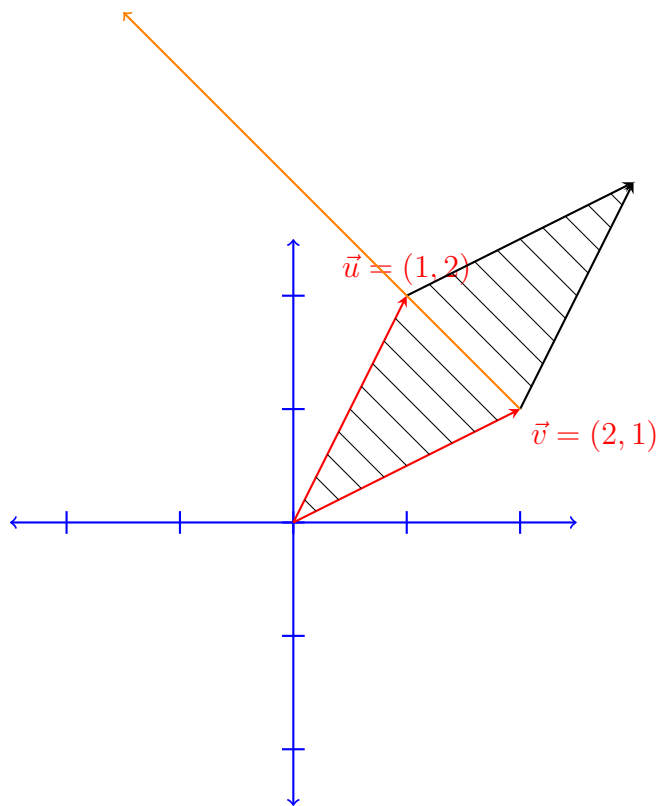
$$c \cdot \vec{u} + (1 - c) \cdot \vec{v} = \vec{v} + c \cdot (\vec{u} - \vec{v}).$$

2. This is the black striped region (including the boundaries). This is a *parallelogram*.
3. Extend the orange ray to a full line, and take the entire region to the southwest of this line (including the line). This is called a *halfspace*. One way to see this is to understand that the points where $a + b = 1$ is going to be the boundary of this region. This set is the full straight line which contains the half ray discussed in the first part. This line splits the plane into two halves and we have to work out which half is the one described by the formula. To work this out note that the case $a = 0$ and $b = 0$ satisfies $a + b \leq 1$. So the corresponding point, which is the zero vector, lies inside the required region. We deduce that we want to take the half-plane to the south-west of the straight line.

Another hint for understanding this region is to rewrite this expression in terms of different vectors:

$$\vec{p} = \frac{1}{2} (\vec{u} + \vec{v}), \quad \vec{q} = \frac{1}{2} (\vec{u} - \vec{v}).$$

What do these vectors look like on the diagram? Try rewriting the formulae with respect to these vectors and see how it looks.



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Problem 8:

Consider an n -sided polygon in the plane $\mathbb{R}^2$ with vertices at points p_1 up to p_n in $\mathbb{R}^2$.

Now we define vectors $\vec{v}_1, \dots, \vec{v}_n$ which go along the edges of this polygon. To be precise, if $1 \leq i < n$, let the vector $\vec{v}_i$ be the vector which goes from the point p_i to the point p_{i+1} . And if $i = n$, let $\vec{v}_n$ be the vector which goes from the point p_n to the point p_1 .

Draw an example of such a polygon, and label it with these points and vectors.

Show that

$$\sum_{i=1}^n \vec{v}_i = \vec{0}.$$

Proof

Let the vector which points from the origin to the point p_i be denoted $\vec{q}_i$.

We can express the edge vectors $\vec{v}_i$ using these vectors as follows:

$$\vec{v}_i = \begin{cases} \vec{v}_i = \vec{q}_{i+1} - \vec{q}_i & \text{if } 1 \leq i < n \\ \vec{v}_i = \vec{q}_1 - \vec{q}_n & \text{if } i = n. \end{cases}$$

Now we check the sum:

$$\begin{aligned} \sum_{i=1}^n \vec{v}_i &= \sum_{i=1}^{n-1} \vec{v}_i + \vec{v}_n \\ &= \sum_{i=1}^{n-1} (\vec{q}_{i+1} - \vec{q}_i) + (\vec{q}_1 - \vec{q}_n) \\ &= \sum_{i=1}^{n-1} \vec{q}_{i+1} - \sum_{i=1}^{n-1} \vec{q}_i + \vec{q}_1 - \vec{q}_n \\ &= (\vec{q}_2 + \dots + \vec{q}_n) - (\vec{q}_1 + \dots + \vec{q}_{n-1}) + (\vec{q}_1 - \vec{q}_n) \\ &= \vec{0}. \end{aligned}$$

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It is also pretty easy to see this fact intuitively using the geometric definition of the sum of two vectors. Try it on the example you drew.

Problem 9:

Let $S \subset \mathbb{R}^n$ denote a set of vectors in $\mathbb{R}^n$.

- (i) Let $\vec{w} \in \mathbb{R}^n$ be a fixed vector. Define a new set S' from S by:

$$S' = \{\vec{v} + \vec{w}, \vec{v} \in S\} \subset \mathbb{R}^n.$$

How is the set of geometric points in $\mathbb{R}^n$ corresponding to S' geometrically related to the set of points in $\mathbb{R}^n$ corresponding to S ?

- (ii) Let $\lambda \in \mathbb{R}$ be a fixed real number and assume $\lambda > 0$. Define a new set S'' from S by:

$$S'' = \{\lambda\vec{v}, \vec{v} \in S\} \subset \mathbb{R}^n.$$

How is the set of geometric points in $\mathbb{R}^n$ corresponding to S'' geometrically related to the set of points in $\mathbb{R}^n$ corresponding to S ?

Answers.

The set of points corresponding to S' is obtained from the set of points corresponding to S by translating it by the vector $\vec{w}$. In other words, every point in the set S is pushed a distance $\|\vec{w}\|$ (this notation refers to the length of the vector $\vec{w}$) in the direction of $\vec{w}$.

The set of points corresponding to S'' is obtained from the set of points corresponding to S by dilating it by a factor λ . In other words, every point corresponding to a vector $\vec{v}$ of the set S is replaced by a point with the same direction to the origin but whose distance to the origin has been multiplied by the factor λ .

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